Name Babar Zareen
Sap ID 74760
subject oop
<u>Lab task Inheritance</u>

Question No = 01

Employee Manager (Inheritance)

CODE:

```java
class Employee {    // name babar zareen sap id 74760
    String name = "Ali";
    int salary = 50000;

    void displayInfo() {
        System.out.println("Name: " + name);
        System.out.println("Salary: " + salary);
    }
}

class Manager extends Employee {
    int bonus = 10000;

    void displayInfo() {
        super.displayInfo(); // name babar zareen sap id 74760
        System.out.println("Bonus: " + bonus);
    }
}

public class Main {
    public static void main(String[] args) {
        Manager m = new Manager();
        m.displayInfo();
    }
}
```

OUTPUT:

```
Salary: 50000
Bonus: 10000


...Program finished with exit code 0
Press ENTER to exit console.
```

Question No = 02

Vehicle Car & Bike (Method Overriding)

CODE:

Main.java

```java
class Vehicle {     //name babar zareen sap id 74760
    void start() {
        System.out.println("Vehicle is starting...");
    }
}

class Car extends Vehicle {
    void start() {
        System.out.println("Car starts with key");
    }
}

class Bike extends Vehicle { //name babar zareen sap id 74760
    void start() {
        System.out.println("Bike starts with kick");
    }
}
public class Main {
    public static void main(String[] args) {
        Car c = new Car();
        Bike b = new Bike();
        c.start();
        b.start();
    }
}
```

OUTPUT:

```
Car starts with key
Bike starts with kick


...Program finished with exit code 0
Press ENTER to exit console.
```

Question No = 03

Person Student (Code Reuse)

CODE:

Main.java

```java
class Person {
    String name = "Babar Zareen";

    void display() {
        System.out.println("Name: " + name);
    }
}

class Student extends Person {
    int rollNo = 74760;

    void displayStudent() {
        display(); // name babar zareen sap id 74760
        System.out.println("Roll No: " + rollNo);
    }
}

public class Main {
    public static void main(String[] args) {
        Student s = new Student();
        s.displayStudent();
    }
}
```

OUTPUT:

```
Name: Babar Zareen
Roll No: 74760


...Program finished with exit code 0
Press ENTER to exit console.
```